1. Hypothesis Testing for Battery Life

Problem:

A battery company claims that its batteries last **50 hours on average**. A consumer group tests **30 batteries** and finds an average of **48 hours**, with a standard deviation of **5 hours**. Test at $\alpha = 0.05$ if the company's claim is true.

Step 1: Define Hypotheses

- **Null Hypothesis (H_0)**: $\mu = 50$ (The battery life is 50 hours)
- **Alternative Hypothesis (H_a)**: $\mu \neq 50$ (The battery life is different)

Step 2: Choose the Test

- **Since $n = 30$ is small** and **σ is unknown**, we use a **T-Test**.

Step 3: Compute the Test Statistic

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$t = \frac{48 - 50}{5/\sqrt{30}}$$

$$t = \frac{-2}{5/5.477} = \frac{-2}{0.912} = -2.19$$

Step 4: Find the Critical Value

- **Degrees of Freedom (df) =** $n - 1 = 30 - 1 = 29$.
- From the **T-table**, the **critical t-value at $\alpha = 0.05$ (two-tailed test)** is ± 2.045.

Step 5: Decision Rule

- Since $|t| = 2.19 > 2.045$, we **reject** H_0.

Conclusion:

✅ The battery life **is significantly different from 50 hours**.

---

2. Hypothesis Testing for GPA

Problem:

A university claims that the **average GPA** of its students is **3.5**. A researcher collects **40 student GPAs** and finds an average of **3.3**, with a standard deviation of **0.6**. Test at $\alpha = 0.01$.

Solution:

- $H_0 : \mu = 3.5$
- $H_a : \mu \neq 3.5$

Since $n = 40$ **(large sample)**, we use a **Z-Test**.

$$Z = \frac{3.3 - 3.5}{0.6/\sqrt{40}}$$

$$Z = \frac{-0.2}{0.0949} = -2.11$$

- **Critical value (Z-table,** $\alpha = 0.01$**)** = ± 2.576.

Since $|Z| = 2.11 < 2.576$, we **fail to reject** H_0.

Conclusion:

✅ There is **not enough evidence** to say the university's claim is false.

---

3. Hypothesis Testing for Blood Pressure Reduction

Problem:

A new drug is supposed to **lower blood pressure by 10 mmHg**. A study on **25 patients** shows an average reduction of **8.5 mmHg**, with a standard deviation of **3.2 mmHg**. Test at $\alpha = 0.05$.

Solution:

- $H_0 : \mu = 10$
- $H_a : \mu \neq 10$

Since $n = 25$ (small sample), we use a **T-Test**.

$$t = \frac{8.5 - 10}{3.2/\sqrt{25}}$$

$$t = \frac{-1.5}{0.64} = -2.34$$

- **Critical value (T-table, $df = 24, \alpha = 0.05$) =** ± 2.064.

Since $|t| = 2.34 > 2.064$, we **reject** H_0.

Conclusion:

✅ The drug **does not lower blood pressure as claimed**.

---

4. Hypothesis Testing for Call Center Wait Time

Problem:

A call center manager claims that the **average customer wait time** is **less than 5 minutes**. A random sample of **20 calls** shows an average wait time of **4.5 minutes**, with a standard deviation of **1.2 minutes**. Test at $\alpha = 0.05$.

Solution:

- $H_0 : \mu \geq 5$
- $H_a : \mu < 5$ **(one-tailed test)**

Using a **T-Test**:

$$t = \frac{4.5 - 5}{1.2/\sqrt{20}}$$

$$t = \frac{-0.5}{0.268} = -1.87$$

- **Critical value (T-table, $df = 19, \alpha = 0.05$, one-tailed) =** -1.729.

Since $-1.87 < -1.729$, we **reject** H_0.

Conclusion:

✅ Customers **wait significantly less than 5 minutes**.

---

5. Hypothesis Testing for Sports Drink Effect

Problem:

A sports drink company claims that its new **energy drink improves running speed**. A test on **15 athletes** shows an **average speed increase of 0.8 seconds**, with a **standard deviation of 0.3 seconds**. Test at $\alpha = 0.01$.

Solution:

- $H_0 : \mu = 0$
- $H_a : \mu > 0$ **(one-tailed test)**

Using **T-Test**:

$$t = \frac{0.8 - 0}{0.3/\sqrt{15}}$$

$$t = \frac{0.8}{0.0775} = 10.32$$

- **Critical value (T-table, $df = 14$, $\alpha = 0.01$, one-tailed) = 2.624**.

Since $t = 10.32 > 2.624$, we **reject** H_0.

Conclusion:

✅ The **sports drink significantly improves running speed**.